Exercise 3: Correlation filter tracking

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I. Introduction

In this assignment we implemented a correlation filter tracker and evaluated its performance on the VOT2014 dataset. The tracker operates by learning a filter in the Fourier domain that produces a Gaussian peak response at the target's location. The filter is then updated over time using a learning rate to adapt to changes in the target's appearance and location. We then evaluated the tracker's performance using VOT-lite toolkit that uses standard performance measures accuracy and robustness.

II. Experiments

The tracker was implemented using a simplified MOSS approach.

A. Analysis of parameters

B. Larger patch

C. Tracking speed

III. Conclusion